Z. Anat. Entwickl.-Gesch. 144, 267—302 (1974)

A Cytoarchitectonic Volumetric Comparison of Brains in Wild and Domestic Sheep*

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Received May 25, 1974

Summary. 1. By means of allometric methods it is possible to determine the intraspecific relation between brain weight and body weight in wild European sheep (*Ovis ammon musimon*) and two breeds of domesticated sheep (*Ovis ammon f. aries*). A reduction in total brain weight of 24% is observed which is due to domestication.

2. The following reductions of the five brain subdivisions are determined for domesticated sheep: telencephalon 26%, diencephalon 21%, mesencephalon 22%, cerebellum 16% and medulla oblongata 13%.

3. In the telencephalon the greatest reduction is found in the allocortex (29%), followed by the neocortex (26%) and corpus striatum (21%).

4. The allocortex is subdivided cytoarchitectionically and functionally into olfactory and limbic structures. There is less reduction in olfactory (22%) than in limbic (35%) components. Within the limbic structures the hippocampus with 41% shows the highest rate of reduction.

5. There are racial differences in brain volume by 8% between North German Moorland Sheep (Heidschnucke) and Blackhead Sheep (Schwarzkopffleischschaf). The sizes and volumes of corresponding brain parts of these two breeds did not differ significantly.

Key words: Sheep — Brain — Cytoarchitectonics — Domestication.

A. Introduction

The present paper is intended to provide a contribution to domestication biology. Its aim is to show size changes of the whole brain and its parts in wild sheep (*Ovis ammon musimon* Schreber, 1782) compared with domestic sheep (*Ovis ammon f. aries* Linnaeus, 1758; cf. Bohlken, 1961) by means of volume measurements.

The recent wild sheep have a holarctic distribution. There are different views whether the present 39 subspecies belong to one or two species (Zalkin, 1951; Kesper, 1953; Herre, 1958; Haltenorth, 1963; Uloth, 1966; Teichert, 1968; Thenius, 1969). As an ancestor of domestic sheep only the Eurasian species *Ovis ammon* L. 1758 comes into consideration (Herre, 1958, 1964, 1966a; Herre and Röhrs, 1955, 1971, 1973; Teichert, 1968). Animals of this species have probably been domesticated at different times in South Europe, in the Near and the Middle East (Herre, 1966a; Herre and Röhrs, 1971, 1973). As documented by osteological studies this event which was of considerable importance to man has probably occurred in the Near East about 11000 years ago (Reed, 1962, 1969). This is why

* I want to thank Professor Dr. M. Röhrs and Dr. D. Kruska for their support and advice during the course of the study.

the sheep may well be considered one of the earliest domestic animals (Herre, 1964, 1966a; Herre and Röhrs, 1971, 1973).

Under the conditions of domestication changes of phenotype, structure and function of organs and behavioural changes have occurred. In a direct comparison of domestic animals with individuals of their wild ancestral species changes are clearly visible. Present knowledge on these phenomena has been recently reviewed by Herre and Röhrs (1971, 1973 etc.).

A comparative investigation of the brain where the influence of domestication can be observed very clearly is of particular interest.

There are several early reports on the reduction of brain size found in domestic animals (Darwin, 1868; Klatt, 1912). Changes in macromorphology of the brain—especially a reduction of the telencephalon, and a reduced fissuration of the hemispheres, i.e. shorter and less deep fissures—could be observed in domestic animals (Klatt, 1921; Rawiel, 1939, Stephan, 1951; Herre, 1955; Schumacher, 1963; Kruska, 1970). By the application of allometric methods (Frick, 1957, 1961, 1969; Rempe, 1962, Röhrs, 1959, 1961) which analyze the influence of body weight on organ weight, it was found that all domestic animals have to a varying degree undergone reductions of brain weight due to domestication (Herre and Röhrs, 1971, 1973).

Evaluations of cranial capacities and brain weights by Klatt (1912) and Stephan (1951) have shown brain reductions of various breeds of domestic sheep compared with mufflons. As these authors used different parameters their results can only be compared indirectly.

The intraspecific-allometric relation between the size of the total brain of its subdivisions and the body weight has proved most suitable for a comparison of brain sizes of wild and domestic animals (Herre and Röhrs, 1971, 1973). Using this parameter and quantitative and cytoarchitectonic methods the effects of domestication on the brain volume and the volume of its subdivisions in domestic sheep in relation to wild sheep have not been reported so far.

In addition to this the results are intended to provide indications as to which central nervous functions have been reduced, and to which extent they may be compared to other results from literature. It is furthermore interesting to find out if it is possible to prove the existence of racial differences observed in domestic sheep analyzed in this paper, by means of a volume comparison based on allometric calculations.

B. Material and Methods

The brains of wild sheep studied here came from populations situated near Lainz/Wien, Springe and Obernkirchen/Hannover[1]. Small flight distances, characteristic bleating, flat, narrow hemispheres with pronounced fissures and brain weights which approximately correspond to those of the North German Moorland Sheep, observed in the "mufflons" from the Obernkirchen region could be interpreted as a proof for crossbreeding with domestic sheep breeds. For this reason only the brains of the sheep from Lainz and Springe were utilized for the calculations.

1 I should like to express my thanks to the forest administration Lainz/Wien and to Oberforstmeister Dr. Meyer-Brenken (Obernkirchen) for the shooting of mufflons, and to the LFZ Hannover and Herrn Meyer (Behringen) for procuring sheep brains, and to Dr. H. Stephan (Frankfurt) who kindly let me have weight data of sheep.

Brains of North German Moorland Sheep coming from the "Lüneburger Heide" and Blackhead Sheep from the region of Lower Saxony have been chosen for comparison with those of the wild sheep.

The age of the animals was determined according to their dentition (Habermehl, 1961).

This study is based on the brain and gross body weight of 6 mufflons, 44 North German Moorland Sheep and 34 Blackhead Sheep (Table 1).

All brains were dissected from the skulls after not more than two hours postmortem, weighed when still fresh, and fixed and preserved in 4% Formaline for about three months. After some swelling the brains reached a constant weight corresponding approximately to the fresh weight (cf. Stephan, 1951). At this moment, after the removal of the Pia mater, photographs of their dorsal, lateral and ventral sides were taken (Table 1).

In accordance with their position relative to the allometric line (see below) the brains of four mufflons (M 1, M 2, M 3, M 4), three North German Moorland Sheep (H 1, H 2, H 3) and three Blackhead Sheep (Sk 1, Sk 2, Sk 3) were embedded into paraffin and cut into frontal serial sections of 20 μ thickness. 250 sections were found to be an appropriate number (Stephan, 1960a; Kruska, 1970). Every third of the working sections which had been coloured in Cresylechtviolett, was photographed on the plate of a light box (Recordak-Film, diaphragm 16, $^1/_2$ sec) and enlarged and printed (Agfa-paper BN 1, 17.8 × 24 cm).

Following Stephan (1954a, b, 1960a, 1964), Kruska (1970) and Kruska and Stephan (1973) the five brain subdivisions medulla oblongata, cerebellum, mesencephalon, diencephalon and telencephalon were circumscribed on the photographs of the serial sections. The telencephalon was further subdivided into neocortex, corpus striatum and allocortex. Moreover in the allocortex the olfactory components—bulbus olfactorius accessorius, regio retrobulbaris, regio praepiriformis regio periamygdalaris+nucleus amygdalae, tuberculum olfactorium, basal nuclei with parts of the commissura anterior—and the limbic structures—septal nuclei, diagonal band of Broca, hippocampus, schizocortex—could be distinguished (Tables 3–5). The defined areas were cut out of the photographs and weighed. The serial sections volume (SSV) was determined according to Stephan (1960a) and the formula:

$$\mathrm{SSV}(\mathrm{mm}^2) = \mathrm{DA}(\mathrm{g})\ \frac{\text{meas. section interval (cm)} \cdot \text{paper value} (\mathrm{cm}^2/\mathrm{g})}{\text{square number of the linear magnification}} \cdot 1000 \qquad (1)$$

DA = defined areas.

Shrinkage caused by the paraffin embedding amounted to approximately 46.5% (44.0–50.6), i.e. the volumes of their serial sections volumes corresponded to only about 53.5% of their original volume.

The fresh volume (total brain volume = TBV) could be calculated from fresh brain weight and the specific weight of brain substance (1.036 g/cm³, Stephan, 1960a):

$$\mathrm{TBV}(\mathrm{mm}^3) = \frac{\text{fresh brain weight (g)} \cdot 1000}{1.036\ (\mathrm{g/cm}^3)} \,. \qquad (2)$$

By multiplying the correcting factor ($k = \mathrm{TBV}/\mathrm{SSV}$) with the volumes of the serial sections the fresh brain volumes could be obtained (Tables 3–4). These values were expressed in percent of the total brain volume (Tables 5–6).

$$\text{Relative Value (RV)} = \frac{\mathrm{BSV}}{\mathrm{TBV}} \cdot 100 \qquad (3)$$

$$\text{Brain Subdivision Volume (BSV)} = \frac{\mathrm{RV} \cdot \mathrm{TBV}}{100} \qquad (3\mathrm{a})$$

For a quantitative determination of brain structures it is essential to choose a parameter which enables us to compare data taken from brains of different dimensions directly with each other. Allometric methods are suitable for intraspecific comparisons of brain sizes of wild and domestic animals (Herre, 1955; Frick, 1957, 1969; Röhrs, 1959; Stephan, 1959, 1960a; Rempe, 1962; Herre and Röhrs, 1971, 1973; Kruska and Stephan, 1973).

Table 1. Brain weight and gross body weight of sheeps

Particulars	No.	BrW (g)	BoW (g)	Sex	Age	Author
O. a. musimon:						
Mufflon	26/66 (M1)	135.0	28500	♀	8–10 Y.	
,,	1262/P(M2)	132.4	30000	♀	6 Y.	
,,	1263/P(M3)	132.5	29000	♀	7 Y.	
,,	1264/P(M4)	132.2	27000	♀	3 Y.	
,,	—	131.0	24000	♀	4–5 Y.	Stephan
,,	—	139.0	30000	♀	—	Brauer u. Schober
O. a. f. aries:						
Heidschnucke	1221/P	120.5	38000	♀	6 Y.	
,,	1222/P	117.9	40000	♀	5 Y.	
,,	1223/P	120.5	42000	♀	6 Y.	
,,	1224/P	107.0	37000	♀	10 Y.	
,,	1225/P	120.0	40000	♀	6 Y.	
,,	1226/P	122.0	39000	♀	10 Y.	
,,	1227/P	106.4	29000	♀	9 Y.	
,,	1228/P	109.6	35000	♀	8 Y.	
,,	1229/P	119.6	35000	♀	6 Y.	
,,	1230/P(H1)	102.4	31000	♀	5 Y.	
,,	1232/P	116.4	36000	♀	9 Y.	
,,	1233/P	118.0	35000	♀	7 Y.	
,,	1234/P	110.0	41000	♀	6 Y.	
,,	1235/P	105.1	36000	♀	10 Y.	
,,	1236/P	111.0	34000	♀	9 Y.	
,,	1237/P	108.2	31000	♀	6 Y.	
,,	1238/P	119.0	42000	♀	7 Y.	
,,	1239/P	112.5	41000	♀	10 Y.	
,,	1240/P	105.7	35000	♀	4 Y.	
,,	1242/P(H2)	109.6	36000	♀	10 Y.	
,,	1297/P	97.3	25000	♂	10 M.	
,,	1298/P(H3)	99.8	24000	♂	10 M.	
,,	1299/P	87.5	26000	♂	10 M.	
,,	1300/P	95.5	25000	♂	10 M.	
,,	1301/P	104.8	26000	♂	10 M.	
Heidschnucke	—	118.0	38000	♀	4–5 Y.	Stephan
,,	—	109.0	31000	♀	8 M.	,,
,,	—	108.0	30000	♀	10 M.	,,
,,	—	106.0	30000	♀	8 M.	,,
,,	—	106.0	42000	♀	4–5 Y.	,,
,,	—	105.0	40000	♀	4–5 Y.	,,
,,	—	105.0	25000	♀	8 M.	,,
,,	—	104.0	38000	♀	4–5 Y.	,,
,,	—	103.0	24000	♀	1 Y.	,,
,,	—	101.0	27000	♀	10 M.	,,
,,	—	96.0	19000	K	11 M.	,,
,,	—	95.0	20000	K	11 M.	,,
,,	—	95.0	20000	K	11 M.	,,
,,	—	94.0	20000	K	11 M.	,,
,,	—	91.0	30000	♀	10 M.	,,
,,	—	109.0	24000	♀	7 M.	,,
,,	—	110.0	28500	♀	7 M.	,,
,,	—	114.0	43000	♀	4–5 Y.	,,
,,	—	108.0	27500	K	10 M.	,,

Table 1 continued

Particulars	No.	BrW (g)	BoW (g)	Sex	Age	Author
Fleischschaf	901/P	130.0	55000	♂	1 Y.	
,,	902/P	122.0	62500	♂	13 M.	
,,	903/P	117.0	52500	♀	3–4 Y.	
,,	931/P	116.8	58000	♀	8–10 Y.	
,,	952/P	117.1	55000	♂	10–12 M.	
,,	950/P	105.1	50000	♂	9 M.	
,,	951/P	100.0	50000	♂	1 Y.	
,,	953/P	122.0	55000	♂	1 Y.	
,,	1001/P(Sk1)	125.0	65000	♀	3–4 Y.	
,,	1002/P	123.0	52000	♀	7 Y.	
,,	1003/P	129.0	60000	♀	10 Y.	
,,	1046/P	101.3	42000	♀	10 M.	
,,	1047/P	102.3	40000	♂	9 M.	
Fleischschaf	1048/P	103.4	45000	♂	9 M.	
,,	1049/P(Sk3)	113.9	43000	♂	1 Y.	
,,	1090/P	115.3	60000	♀	3–4 Y.	
,,	1091/P	115.5	57000	♀	3–4 Y.	
,,	1092/P	113.2	59000	♀	2–3 Y.	
,,	1093/P	138.0	68000	♀	4 Y.	
,,	1105/P	106.4	52000	♂	10 M.	
,,	1106/P(Sk2)	120.0	55000	♂	1 Y.	
,,	1107/P	120.5	50000	♂	1 Y.	
,,	1108/P	120.0	52000	♂	11 M.	
,,	1146/P	121.5	49000	♀	7 Y.	
,,	1147/P	126.5	53000	♀	2 Y.	
,,	1148/P	117.6	59000	♀	6 Y.	
,,	1149/P	110.8	60000	♀	10 Y.	
,,	1150/P	117.4	46000	♀	7 Y.	
,,	—	135.0	64000	K	1–2 Y.	Stephan
,,	—	132.0	83000	♂	1–2 Y.	,,
,,	—	122.0	74500	—	1–2 Y.	,,
,,	—	115.0	61000	K	1–2 Y.	,,
,,	—	115.0	61000	—	1–2 Y.	,,
,,	—	114.0	65500	—	1–2 Y.	,,

As an expression of a consistant relationship between organ weights and body weights Snell (1891) and Dubois (1897), using the brain as an example, defined the size of the organ as a function of body weight (allometry equation):

$$y = b \cdot x^{a}. \tag{4}$$

(y = body weight, x = organ weight, a = allometric exponent, b = organ coefficient.)

Expressed in logarithms this equation describes a straight line:

$$\log \text{BrW} = \log b + a \cdot \log \text{BoW} \tag{4a}$$

$$\log b = \log \text{BrW} - a \cdot \log \text{BoW} \tag{4b}$$

(BrW = brain weight, BoW = body weight.)

In the double logarithmic scale the allometric constant a indicates the tan α of the angle of inclination of the line; it characterizes the dependence of brain weight from body weight. The logarithm of the organ coefficient b determines the point of intersection of the line with the ordinate in $\log x = 0$ and informs thus on the influence of factors which determine the brain weight, independently from the body weight. Hence it follows according to 4b:

$$\log b = \log \text{BrW} \quad (\log x = 0). \tag{4c}$$

Brain and gross body weights of wild and domestic sheep, as well as those of North German Moorland Sheep and Blackhead sheep were plotted in a double logarithmic scale (Table 2 and 9). By regression calculation it was possible to determine separate gradients and organ coefficients of the regression line (RL), of the reduced main axis of the ellipse (RMA), and of the main axis of the ellipse (MAE) (Table 2). The following calculations were carried out by using the coefficient of the MAE in order to create a good basis for comparison to Kruska's investigations. The F-Test (Rempe, 1962, Table 2) was used in order to check the significance of position and inclination of the MAE.

Inclination values of $a=0.20$ to $a=0.30$ are characteristic for intraspecific relations of brain weight to body weight (Röhrs, 1955; Bährens, 1958; Schumacher, 1963; Kruska, 1970; Weidmann, 1970; Ebinger, 1972).

A comparison of brain and body weight in wild and domestic sheep resulted in an intraspecific exponent of 0.20 for domestic sheep. Due to the small number of wild speciments available this exponent could not be proved unequivocally (see quantitative results). This is why the following calculations of brain weight changes will refer to gradient values of $a=0.20$, $a=0.25$ and $a=0.30$. In addition to this the confrontation of these exponents is intended to show how much the changes of brain weight depend on the allometric exponents (Table 8).

Under these conditions the following allometric equations can be established for wild and domestic sheep:

$$\log \overline{\mathrm{BrW}}_{\mathrm{wild}} = \log b + 0.20 \cdot \log \overline{\mathrm{BoW}}_{\mathrm{wild}} \qquad (5)$$
$$\log \overline{\mathrm{BrW}}_{\mathrm{dom.}} = \log b + 0.20 \cdot \log \overline{\mathrm{BoW}}_{\mathrm{dom.}}$$

$$\log \overline{\mathrm{BrW}}_{\mathrm{wild}} = \log b + 0.25 \cdot \log \overline{\mathrm{BoW}}_{\mathrm{wild}} \qquad (6)$$
$$\log \overline{\mathrm{BrW}}_{\mathrm{dom.}} = \log b + 0.25 \cdot \log \overline{\mathrm{BoW}}_{\mathrm{dom.}}$$

$$\log \overline{\mathrm{BrW}}_{\mathrm{wild}} = \log b + 0.30 \cdot \log \overline{\mathrm{BoW}}_{\mathrm{wild}} \qquad (7)$$
$$\log \overline{\mathrm{BrW}}_{\mathrm{dom.}} = \log b + 0.30 \cdot \log \overline{\mathrm{BoW}}_{\mathrm{dom.}}$$

(log BrW; log BoW = arithmetic mean of the logarithms of brain of body weights respectively.)

A comparison of organ coefficients informs on the extent of brain weight changes in wild and domestic sheep. A higher b-value of the wild compared to the domestic line indicates the reduction of brain weight during domestication. The following calculations can be carried out (modified, according to Kruska and Stephan, 1973):

$$\text{Total Brain Reduction (TBR)} = \frac{b_{\mathrm{dom.}}}{b_{\mathrm{wild}}} \cdot 100 - 100. \qquad (8)$$

According to (4c) or in the case of the same body weight of wild and domestic animals follows:

$$\mathrm{TBR} = \frac{\mathrm{BrW}_{\mathrm{dom.}}}{\mathrm{BrW}_{\mathrm{wild}}} \cdot 100 - 100 \quad (\mathrm{BoW}_{\mathrm{dom.}} = \mathrm{BoW}_{\mathrm{wild}}). \qquad (9)$$

Values of brain weight were converted into values of brain volume according to (2):

$$\mathrm{TBR} = \frac{\mathrm{TBV}_{\mathrm{dom.}}}{\mathrm{TBV}_{\mathrm{wild}}} \cdot 100 - 100. \qquad (9\mathrm{a})$$

According to this the reduction of brain subdivisions (BSR) can be calculated as follows:

$$\mathrm{BSR} = \frac{\mathrm{BSV}_{\mathrm{dom.}}}{\mathrm{BSV}_{\mathrm{wild}}} \cdot 100 - 100. \qquad (9\mathrm{b})$$

By inserting (3a) into (9b) one obtains:

$$\mathrm{BSR} = \frac{\mathrm{RV}_{\mathrm{dom.}} \cdot \mathrm{TBV}_{\mathrm{dom.}}}{\mathrm{RV}_{\mathrm{wild}} \cdot \mathrm{TBV}_{\mathrm{wild}}} \cdot 100 - 100. \qquad (10)$$

According to (9a) we can say that

$$\frac{\mathrm{BRV}_{\mathrm{dom.}}}{\mathrm{BRV}_{\mathrm{wild}}} \cdot 100 = \mathrm{TBR} + 100.$$

Hence the reduction in percent of a brain subdivision is calculated according to:

$$\mathrm{BSR} = \frac{\mathrm{RV}_{\mathrm{dom.}}}{\mathrm{RV}_{\mathrm{wild}}} \cdot (\mathrm{TBR} + 100) - 100. \qquad (10\mathrm{a})$$

In order to evaluate the degree of reduction of the brain subdivisions compared with the total volume reduction, Herre and Röhrs (1973) determined the total reduction as 100%. For the following course of calculation this means:

$$\text{Percentage of BS in TBR} = \frac{\mathrm{BSV}_{\mathrm{wild}} \cdot \mathrm{BSR}}{\mathrm{TBV}_{\mathrm{wild}} \cdot \mathrm{TBR}} \cdot 100. \qquad (11)$$

Inserting (3) into (11) leads to:

$$\text{Percentage of BS in TBR} = \mathrm{RV}_{\mathrm{wild}} \cdot \frac{\mathrm{BSR}}{\mathrm{TBR}} \,. \qquad (11\mathrm{a})$$

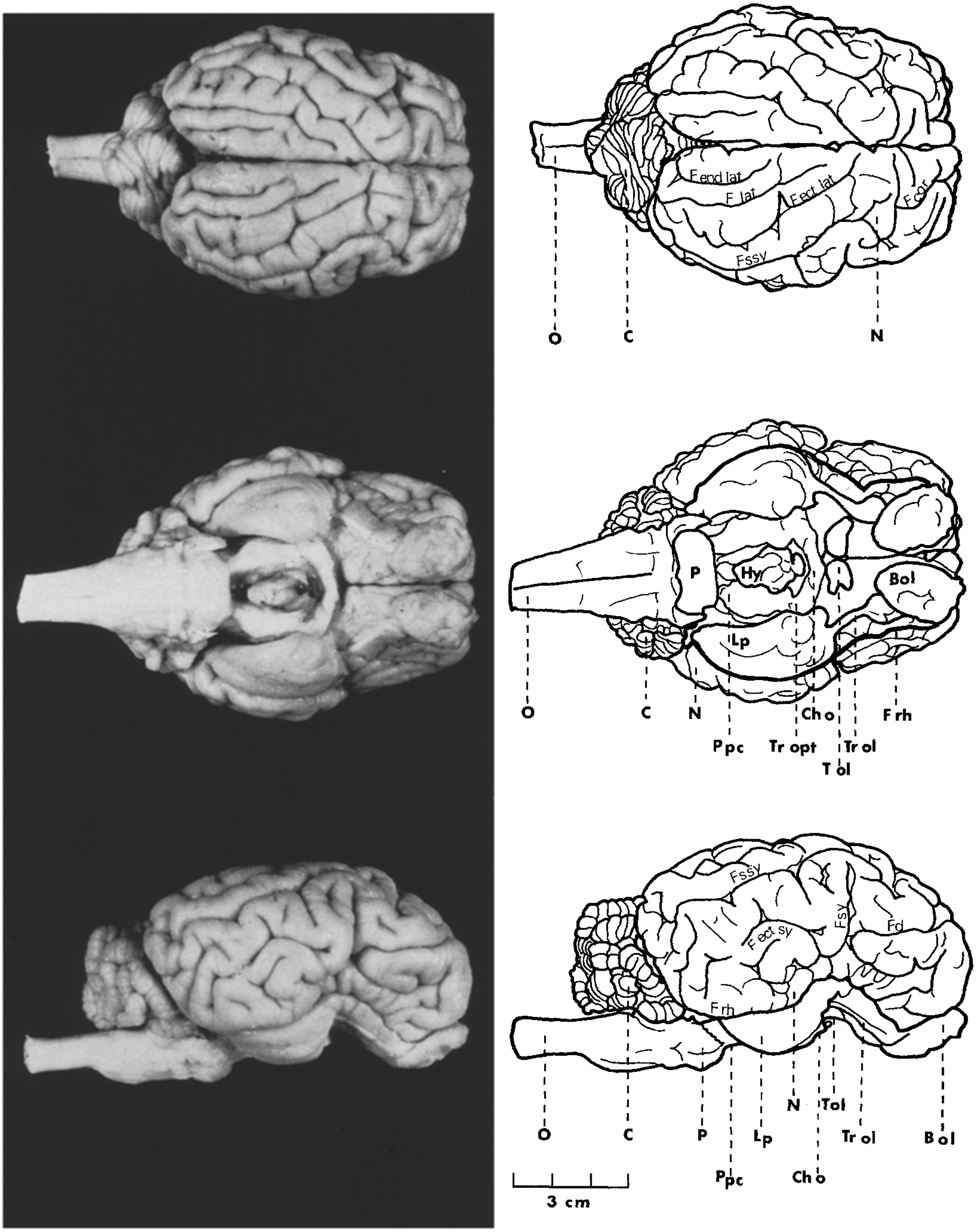
The allometric calculations are based on the logarithms of the single values. According to Rempe (1962) logarithms of body and organ measurements give a more even distribution than the values themselves. This is why even in a univariate comparison of brain volumes geometrical means should be used. The geometrical variability coefficient (GVC) could be used as a dispersion rate (cf. Helmuth and Rempe, 1968; Kruska and Stephan, 1973; Weber, 1967).

The reduction values determined for the total brain weight of sheep correspond to the reduction rates of the total brain volume. For the determination of the relative values, however, the "pure" brain volume was taken as a reference. This means: As during preparation brain nerves and spinal cord are not always separated in the same length, these varying amounts of tissue were measured separately as "brain residues". These values and the data for the ventricles showing irregular deformations were excluded from the total calculations. Thus the "pure brain volume" as defined here means total brain volume minus the volumes of remaining parts and ventricles. The reduction rate in the pure brain volume is therefore about 1% higher than that in the total brain volumes.

A certain factor of instability which can hardly be avoided in quantitative and cyto-architectonic studies of this kind is the clear delimitation of brain areas. The extent to which phylogenetically old parts are penetrated by new ones (Starck, 1965) varies. But as this is almost the same for all percentages it does not effect the relation.

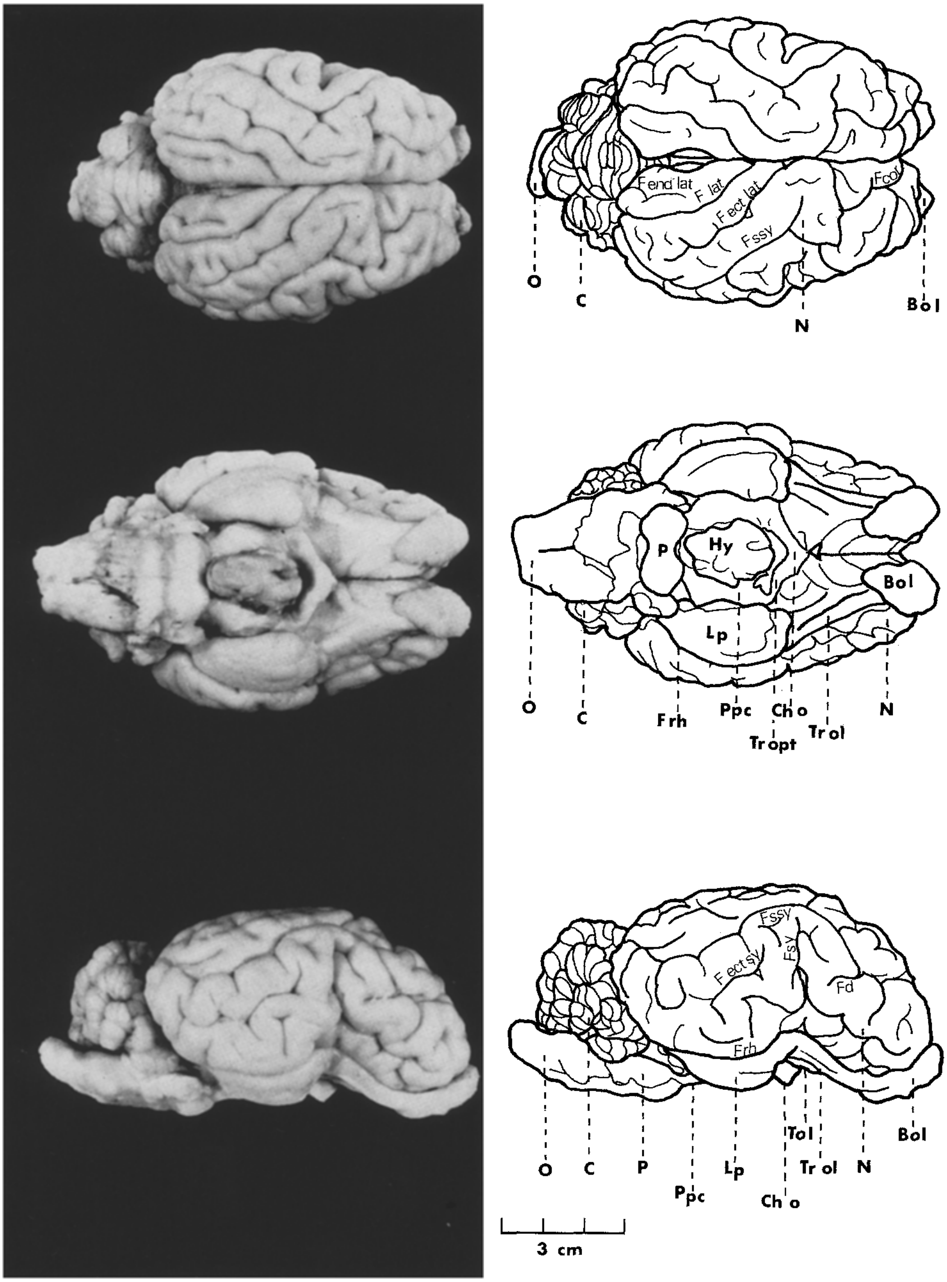
The delimitations for the main regions are demonstrated in Figure 3.

Abbreviations Used in Figs. 1, 3 and 5. Bac = bulbus accessorius; *BN* = basal nuclei; *Bol* = bulbus olfactorius; *C* = cerebellum; *Cho* = chiasma opticum; *D* = diencephalon; *DB* = diagonal band; *Epi* = epiphysis; *Fcor* = fissura coronalis; *Fd* = f. diagonalis; *Fectlat* = f. ectolateralis; *Fectsy* = f. ectosylvia; *Fendlat* = f. endolateralis; *Flat* = f. lateralis; *Frh* = f. rhinalis; *Fssy* = f. suprasylvia; *Fsy* = f. sylvia; *Hi* = hippocampus; *Hy* = hypophysis; *Lp* = lobus piriformis; *M* = mesencephalon; *N* = neocortex; *Na* = nucleus amygdalae; *O* = medulla oblongata; *P* = pons; *Pam* = regio periamygdalaris; *Ppc* = pes pedunculi cerebri; *Prpi* = regio praepiriformis; *Rb* = regio retrobulbaris; *S* = septal nuclei; *Sc* = spinal cord; *Sch* = schizocortex; *St* = corpus striatum; *Tol* = tuberculum olfactorium; *Trol* = tractus olfactorius; *Tropt* = tractus opticus; *V* = ventricle.



Ovis ammon musimon M 3

Fig. 1. Dorsal, ventral and lateral views of one sheep brain each with similar body weight—left



Ovis ammon f. aries H 4

Ovis ammon musimon (27 kg) right Ovis ammon f. aries (29 kg). For abbreviations see p. 273.

C. Quantitative Results

I. Total Brain

The brain of mammals, serving as a centre of integration, association and coordination, has a certain "basic" size which enables the innervation of the different organ systems. This "basic" size depends on several factors (Starck, 1965; Stephan and Bauchot, 1965):

1. Body Size

A change of body size is correlated with a change of brain size, i.e. there is a correlation depending on sizes between the mass of the body which has to be innervated and the brain tissue.

2. Evolution

Development and differentiation of higher centres (e.g. neocortex and neocerebellum) are linked with the penetration of "new" components in phylogenetically old brain regions. This so-called Neencephalization (Starck, 1965) may be considered as an evolutionary index.

3. Specialization

A specifically functional adaptation of animals to certain environments and living conditions implies a special development of corresponding parts in the brain or vice versa. In some aquatic mammals (dolphin, whale), for example, the olfactory centres have been widely reduced. Specializations have to be judged particularly as to their development of sensory systems and the corresponding centres in the brain (Starck, 1965).

The influence of body size may be determined by using the allometry formula. In interspecific comparisons between mammals of closely related species an a-value of about 0.56 (0.5–0.6) has been found so far. Intraspecific comparisons, on the contrary, i.e. comparisons within one species, showed gradient values of the allometric line of about $a = 0.25$ (0.2–0.3).

If the interspecific comparison is intended to provide informations on the specialization and development of higher centres, the influence of body size has to be inserted as $a = 0.56$ (0.5–0.6). If, however, in an intraspecific comparison the effects of different factors on the brain and its parts are to be tested, the influence of body size has to be inserted as $a = 0.25$ (0.2–0.3).

In a comparison of brain sizes of wild and domestic sheep one has to depart from intraspecific allometric relations, from which there may be drawn conclusions on the influence of domestication on brain size.

For the relation brain weight to gross body weight there have been determined separate allometric lines for wild and domestic sheep (Table 2). The few data on wild animals ($n = 6$) and the small dispersion of body weights cause an inclination of the mufflon line (main axis of ellipse) of only 0.13 ($P > 0.05$). For domestic sheep ($n = 78$) the calculation of the gradient yielded the exponent $a = 0.20$ ($P < 0.01$; Linder, 1960; Table 5). It is insufficient to use the significance level of $P > 0.05$ of the wild sheep line as the only basis for the calculation of a

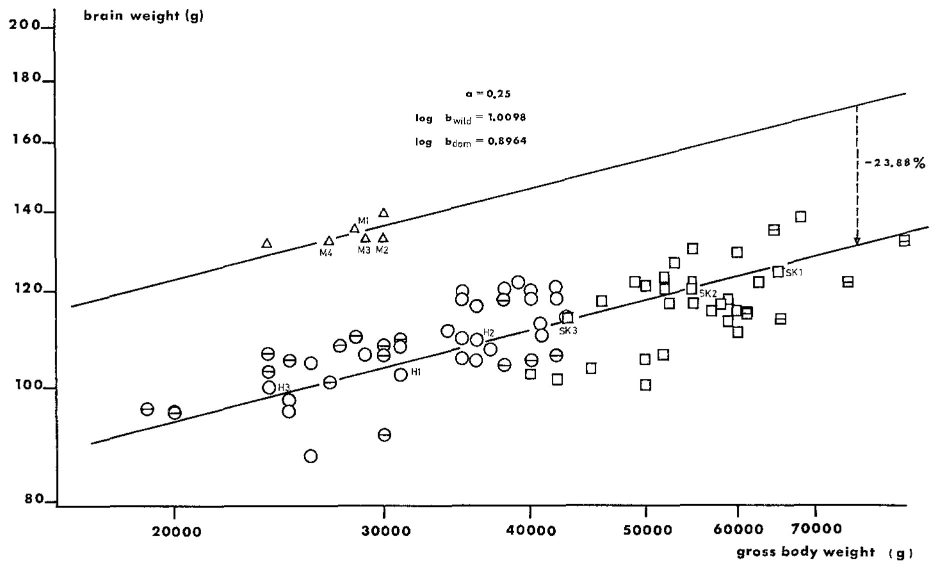


Fig. 2. Relations between brain weight and gross body weight of wild and domestic sheep in a double logarithmic scale. △ mufflons; ○ North German Moorland Sheep (own data); □ Blackhead Sheep; ⊖, ⊟ data given by Stephan; *M, H, Sk* brains studied cytoarchitectionically

Table 2. Statistical values for the relations log brain weight (y) and log gross body weight (x) of wild and domestic sheep

	Mufflon		Domestic sheep	Moorland sheep		Blackhead sheep
N	6		78	44		34
$\bar{x}$	4.4472		4.6050	4.4966		4.7453
$\bar{y}$	2.1260		2.0470	2.0294		2.0697
a_{RL}	0.1493		0.1990	0.2591		0.3106
a_{RMA}	0.2530		0.2653	0.3520		0.5061
a_{MAE}	0.1259		0.2050	0.2736		0.3626
r	0.7192		0.7500	0.7360		0.6138
$T^{\mathrm{F}}\ ^{\mathrm{P}}0.05^{=3.97}$		47.0086			5.3273	
i^{a}		0.2048			0.2955	
$i^{\mathrm{r}}\ ^{\mathrm{P}}0.01^{=0.08}$		0.7494			0.6848	
i^{f}		4.8818			3.3843	
$i^{\mathrm{F}}\ ^{\mathrm{P}}0.01^{=7.00}$		0.0106			4.3920	

common allometric exponent for both random tests. Investigations on the relation brain weight to body weight have up to now resulted in parallel allometry lines for wild and domestic animals (Röhrs, 1955; Schumacher, 1963; Kruska, 1970; Weidemann, 1970; Ebinger, 1972). "The intraspecific dependence of brain weight on body weight existing in wild species has subsisted in domestic animals" (Herre and Röhrs, 1971, p. 104). Significant relations between brain and body weight of a given domestic species may hence be transferred to their wild species. Under these conditions the common inclination value of $a = 0.20$ may be considered as sufficiently proved.

The association of the pairs of values with the calculated line in Fig. 2 illustrates again the linear interdependence of brain weight and gross body weight in the double logarithmic scale.

According to the F-Test (Rempe, 1962) the positions of the allometric lines of the domestic and wild sheep differ significantly ($P < 0.01$). The difference of the b-values determines their difference of position and characterizes thus the reduction of brain weight in the domestic state.

With allometric exponents of 0.20, 0.25 and 0.30 the following equations may be established for wild and domestic sheep lines (cf. Material and Methods, p. 272):

$$\begin{array}{lll} \text{(Wild)} & 2.1260 = 1.2366 + 0.20 \cdot 4.4472 & \text{see (5)} \\ \text{(Dom.)} & 2.0470 = 1.1260 + 0.20 \cdot 4.6050 & \\ \text{(Wild)} & 2.1260 = 1.0142 + 0.25 \cdot 4.4472 & \text{see (6)} \\ \text{(Dom.)} & 2.0470 = 0.8957 + 0.25 \cdot 4.6050 & \\ \text{(Wild)} & 2.1260 = 0.7918 + 0.30 \cdot 4.4472 & \text{see (7)} \\ \text{(Dom.)} & 2.0470 = 0.6655 + 0.30 \cdot 4.6050 & \end{array}$$

This results in reduction values of 22.48% ($a = 0.20$), 23.88% ($a = 0.25$) and 25.24% ($a = 0.30$) for the brain weight of domestic sheep compared to that of wild sheep. On an average the brain of domestic sheep is by 24% smaller than in equally sized wild sheep.

Already in 1912 Klatt was able to prove by means of skull capacity measurements a reduction of brain sizes of domestic sheep compared to those of wild sheep. When—using Klatt's data—the third roots of the skull volumes are correlated to the Condylo basal distances plotted in a double logarithmic scale, a regression calculation for domestic sheep shows a reduction of the skull capacity of 22.12% ($a = 0.30$). Hence the reduction value is by 3.12% lower than that which has been calculated using brain weight data. In spite of this different method the data which Klatt provides may serve as a confirmation of the brain weight reductions in domestic sheep found in this study. Hence we may assume that the Condylo basal distance of sheep, too, is a suitable parameter for a comparison of brains.

Reductions of brain weight of the wild animal compared to that of the domestic animal have so far been demonstrated for several species (Herre and Röhrs, 1971, 1973). It was observed that the rate of the brain reduction can vary considerably according as shown below:

Guinea pig = about 5% (Herre and Röhrs, 1973), albino rat = 9% (Stephan, 1954; Ebinger, 1972), domestic rabbit = 9% (Choinowski, 1958), domestic horse and domestic ass = about 16% (Kruska, 1973), llama and alpaca = 19% (according to Herre and Thiede, 1965), domestic cat = 23% (Röhrs, 1955), domestic sheep = 24%, domestic dog (general) = 31% (Herre and Röhrs, 1971, 1973), ferret = 33% (according to Schumacher, 1963), domestic pig = 34% (Kruska, 1970).

It is of special interest in this regard that Nord (1963) and Rohn (1971) could not determine any reduction of brain weight for the albino mouse compared to its wild species.

In general it may well be assumed that brain sizes are reduced in the course of domestication. One question however, should be answered: if these brain reductions in domestication apply to all brain subdivisions in the same way or if there are regional differences. This question will be discussed below, as regards sheep.

II. Volume Comparison of Brain Subdivisions

According to previous investigations size changes of single brain subdivisions do not correspond to the total brain reductions of domestic animals. The size of five brain parts is reduced during domestication, this reduction is not uniform but varies in dimension (Schumacher, 1963; Herre and Thiede, 1965; Kruska, 1970, 1972; Schleifenbaum, 1973; Kruska and Stephan, 1973).

Brain weights, total brain volumes and absolute volumes of the measured brain subdivisions of wild and domestic sheep are given in the Tables 3 and 4. The percentages comparing brain structure to pure brain volume are listed in Table 5 and 6. Geometrical means, geometrical variability coefficients and relative values of body weights, brain weights, brain volumes and volume of brain subdivisions of wild and domestic sheep are indicated in Table 7. The reductions of brain subdivisions (in percent) are to be seen in Table 8 and Fig. 4. Position and extent of brain subdivisions are shown in Fig. 5.

The following reductions which are discussed in the text, too, refer to the allometric exponent $a = 0.25$ which can be considered as a central value for an intraspecific comparison.

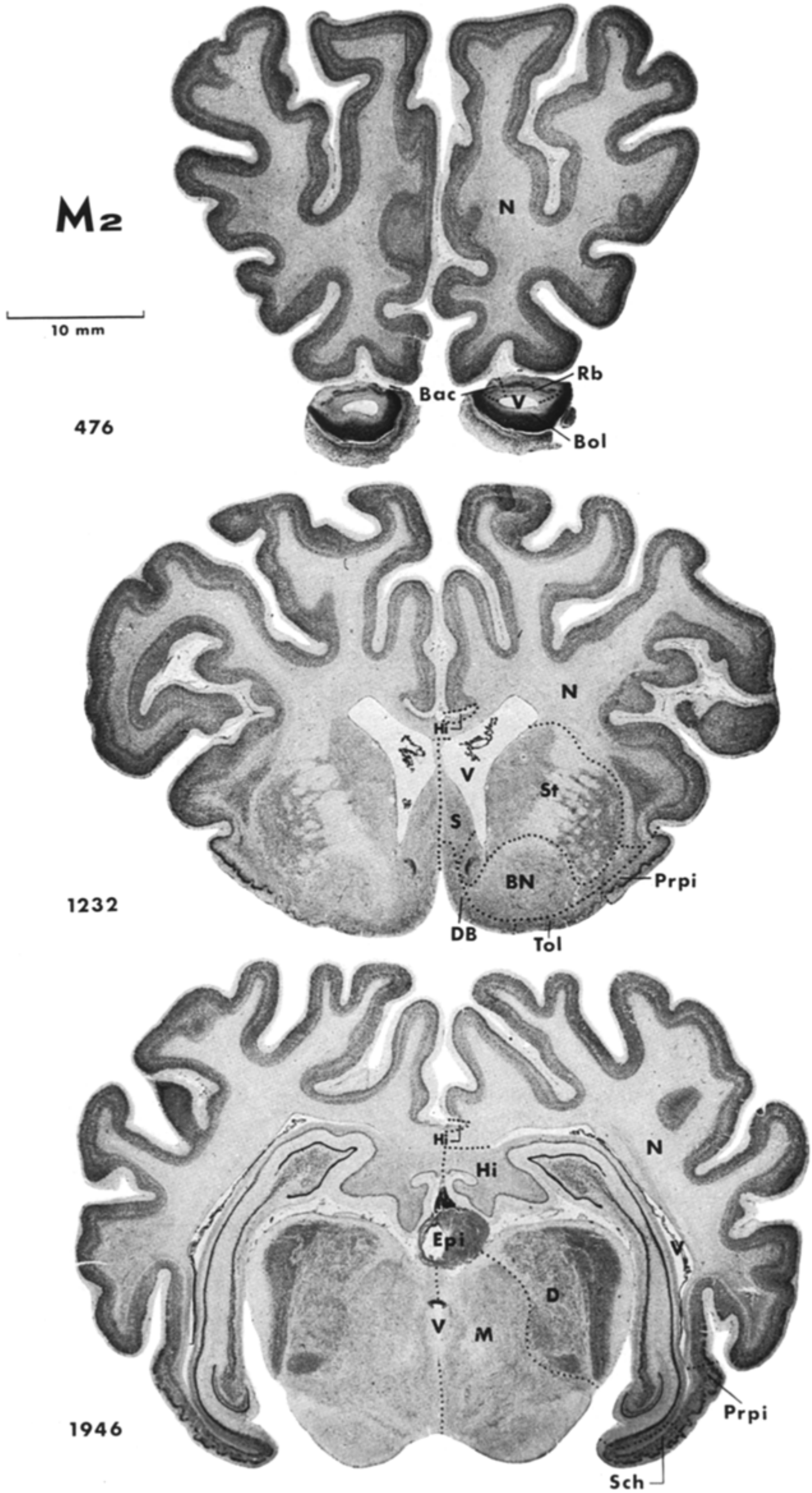
The calculation of the reduction of the five brain subdivisions of domestic sheep gave the following results:

Telencephalon 26.30%, diencephalon 21.18%, mesencephalon 21.51%, cerebellum 15.78%, medulla oblongata 13.39%.

The telencephalon underwent the greatest reduction of all brain subdivisions and surpasses the total reduction of 23.9% by 2.4%. Diencephalon and mesencephalon show about the same second-big value. The cerebellum, followed by the medulla, shows the smallest volume change.

When the systematical subdivision of the brain into five parts is replaced by the "genetic" subdivision into two parts (Starck, 1962), then telencephalon and diencephalon are combined and called prosencephalon, while mesencephalon, cerebellum and medulla oblongata are combined and called rhombencephalon. According to this classification the prosencephalon of domestic sheep is reduced much more (25.8%) in volume than the rhombencephalon which is "only" reduced by 16.5%.

Investigations of this kind on brains of domestic animals have so far been undertaken on a small scale only. A comparison of the reductions values determined here with other results of domestic animal investigations provides therefore additional informations for a general statement on the variability of brain during domestication. First, however, we have to give some additional information. The Tables 9 and 10 facilitate a direct comparison of the reduction values of the brains of several domestic animals. For this purpose some results reported in the literature were adapted to the calculation method described here and put into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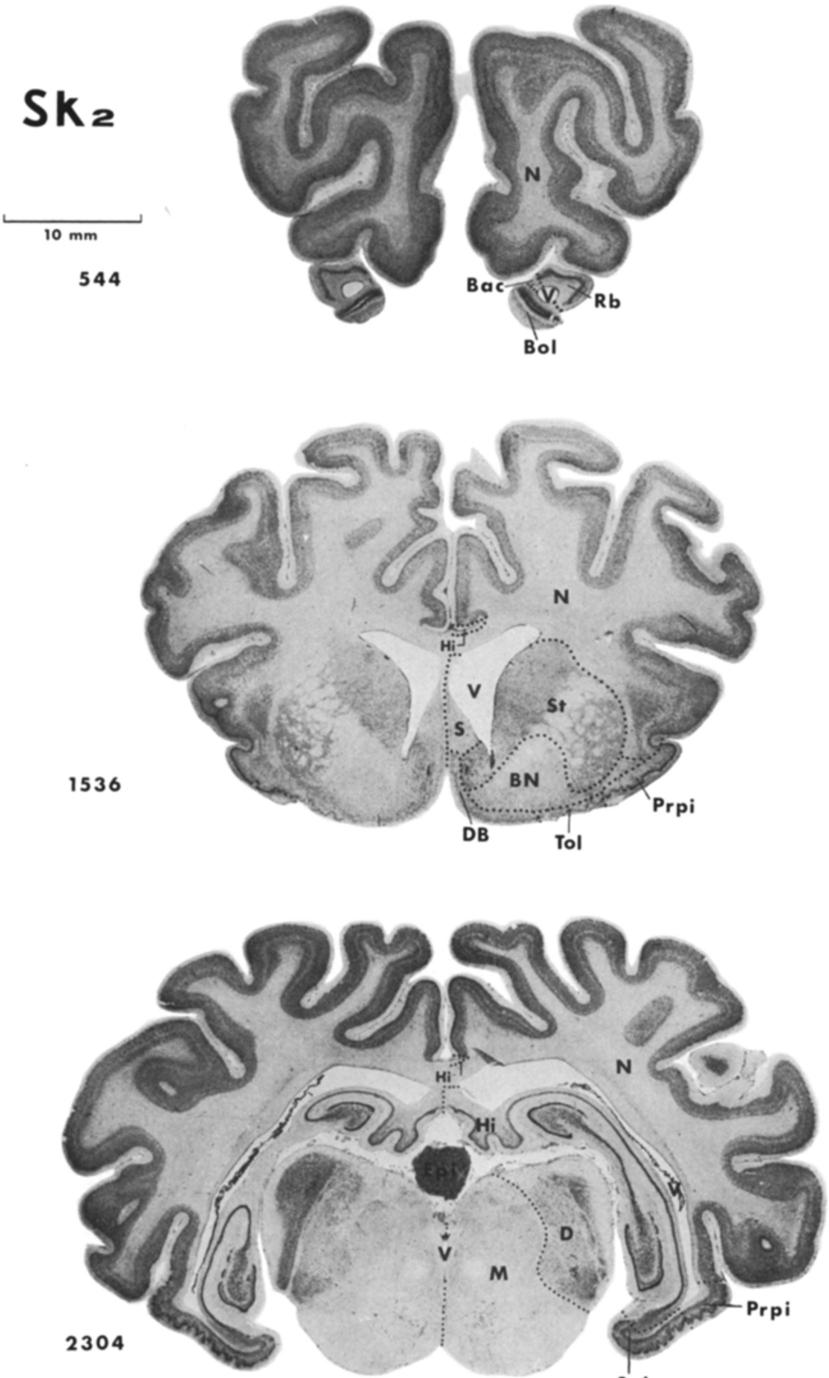

Fig. 3. Cytoarchitectonic delimitation of brain subdivisions on the photographed serial sections of a wild sheep brain (*M 2*) in comparison with a domestic sheep brain (*Sk 2*). For abbreviations see p. 273

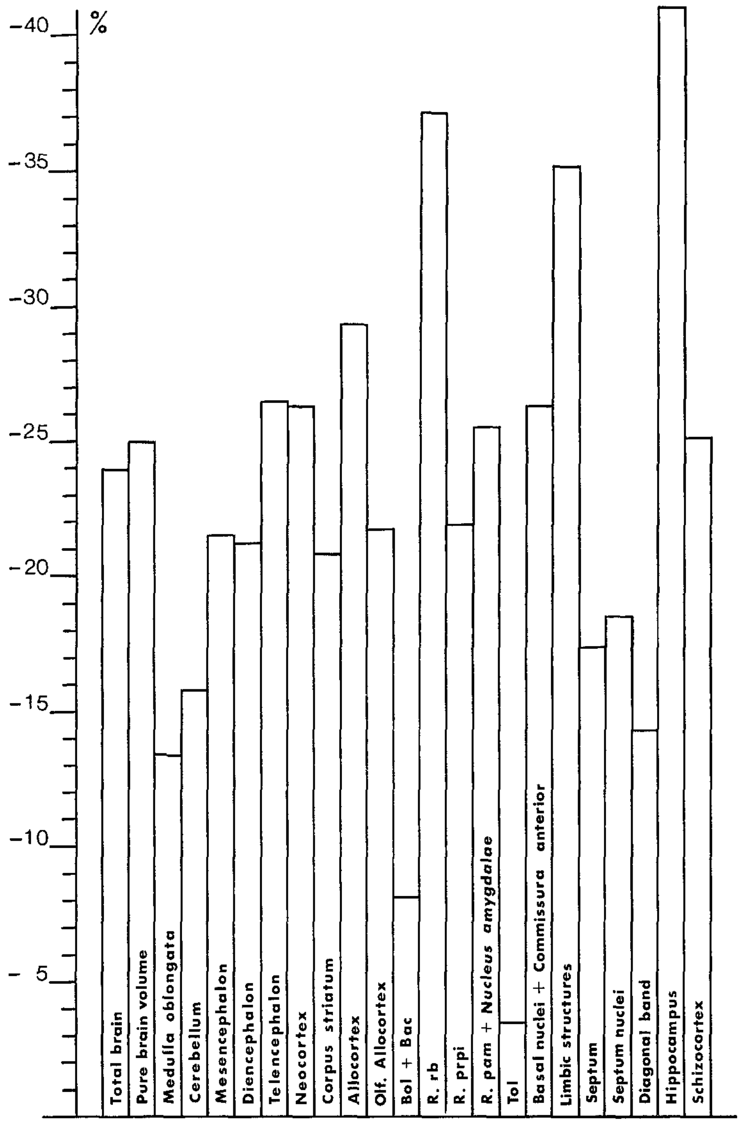

Fig. 4. Diagram of determined reduction values of single brain parts and structures from wild to domestic sheep

relation with the common intraspecific allometric exponent of 0.25. Due to this fact there arise some minor differences compared to the original data.

In the five compared domestic species—albino rat, domestic sheep, domestic pig, ferret and poodle—the telencephalon shows in all cases the greatest reductions. This highly developed part of the mammal central nervous system surpasses thus the value of total reductions.

Table 3. Brain weights (g), total brain volumes (mm³) and absolute volume values (mm³) of the measured brain subdivisions of wild sheep

	M1	M2	M3	M4
Brain weight	135.0	132.4	132.5	132.2
Total brain volume	130308	127799	127896	127606
Remaining parts	1827	2417	2736	2488
Ventricle	2240	3097	3409	2625
Pure brain volume	126242	122286	121751	122493
Medulla oblongata	6501	6689	5677	5588
Cerebellum	13688	13563	12723	13879
Mesencephalon	6442	5865	5745	5579
Diencephalon	8929	8807	8920	8174
Telencephalon	90683	87381	88686	89273
Neocortex	74495	71231	72655	73980
Corpus striatum	4668	4195	4296	4042
Allocortex	11520	11955	11736	11251
Olfactory Allocortex	5044	5185	4936	5186
Bulbus olfactorius + B. olf. accessorius	467	567	564	570
Regio retrobulbaris	218	239	238	267
Regio praepiriformis	1754	1775	1636	1642
Regio periamygdalaris + Nucleus amygdalae	1094	1179	1174	1223
Tuberculum olfactorium	251	307	255	246
Basal nuclei + Commissura anterior	1261	1119	1096	1238
Limbic structures	6476	6770	6773	6066
Septum	764	772	638	693
Septum nuclei	556	568	466	523
Diagonal band	208	205	171	170
Hippocampus	4576	4613	4920	4217
Schizocortex	1136	1385	1216	1156

The other four brain subdivisions, however, cannot be placed into a general order according to the size of their reduction. We can only state that the diencephalon and the cerebellum in most cases show larger reductions than mesencephalon and medulla oblongata. When telencephalon and diencephalon are joined to form the prosencephalon, it is clearly visible that in domestic mammals the forebrain has been reduced considerably.

The data given in Table 9 are overall values which do not provide any information on size relations within the brain. The volume data in the brains of wild and domestic animals can be arranged in the following order (from the largest to the smallest subdivision): telencephalon-cerebellum-diencephalon-medulla oblongata-mesencephalon.

In this respect it is quite interesting to show to what extent the reduced volumes of the brain subdivisions contribute to the total reduction.

Table 4. Brain weights (g), total brain volumes (mm^3) and absolute volume values (mm^3) of the measured brain subdivisions of domestic sheep

	Sk1	Sk2	Sk3	H1	H2	H3
Brain weight	125.0	120.0	113.9	102.4	109.6	99.8
Total brain volume	120656	115830	109942	98842	105792	96361
Remaining parts	4324	3570	3304	3224	2263	2582
Ventricle	3531	2447	2447	2281	2056	2359
Pure brain volume	112802	109813	104191	93338	101473	91420
Medulla oblongata	5830	6752	5248	5505	5953	5247
Cerebellum	13495	13647	12879	11635	11008	11473
Mesencephalon	5084	5365	4748	5385	5182	4479
Diencephalon	7587	7258	7624	7362	8294	6675
Telencephalon	80805	76791	73692	63452	71037	63547
Neocortex	67074	64066	60730	50674	57431	53274
Corpus striatum	3858	3510	3700	3804	4476	2982
Allocortex	9874	9214	9262	8974	9129	7292
Olfactory Allocortex	4944	4472	4369	4288	4362	3651
Bulbus olfactorius + B. olf. accessorius	631	544	577	516	563	424
Regio retrobulbaris	171	150	169	156	211	135
Regio praepiriformis	1762	1507	1563	1330	1270	1288
Regio periamygdalaris + Nucleus amygdalae	1151	1064	988	861	864	786
Tuberculum olfactorium	289	335	287	256	247	255
Basal nuclei + Commissura anterior	940	871	786	1169	1207	764
Limbic structures	4930	4743	4893	4687	4768	3640
Septum	715	603	604	674	671	595
Septum nuclei	541	439	444	505	472	409
Diagonal band	174	164	160	169	199	187
Hippocampus	3165	3022	3029	3115	3153	2278
Schizocortex	1050	1117	1260	898	944	767

From the comparison of the data in Table 10 it may be seen that the reduction is most pronounced in the telencephalon. In the domestic sheep, the domestic pig, the ferret and the poodle the participation of the telencephalon in the total reduction (80%) is larger than that found in the white rat (70.9%).

In this paper medulla oblongata, cerebellum, mesencephalon and diencephalon of the sheep brains were not further subdivided according to their function, so that only very general conclusions can be drawn from the overall reductions in the main subdivisions.

Decrease of vision during domestication indicated by reductions of receptor cells in the retina and reductions of optic centres in the brain of domestic pigs are reported by Wigger (1939), Stephan (1951), and Kruska (1972). Changes in the stato-acoustic system are discussed as well for domestic animals (Herre and Röhrs, 1971, 1973).

Table 5. Relative percentages (RV) of measured brain structures of the wild sheep brains. Pure brain volume = 100%

	M1	M2	M3	M4
Medulla oblongata	5.1496	5.4536	4.6628	4.5619
Cerebellum	10.8427	11.0912	10.4500	11.3304
Mesencephalon	5.1029	4.7961	4.7186	4.5545
Diencephalon	7.0729	7.2020	7.3264	6.6730
Telencephalon	71.8327	71.4563	72.8421	72.8801
Neocortex	59.0097	58.2495	59.6751	60.3953
Corpus striatum	3.6977	3.4305	3.5285	3.2998
Allocortex	9.1253	9.7763	9.6393	9.1850
Olfactory Allocortex	3.9955	4.2401	4.0764	4.2337
Bulbus olfactorius + B. olf. accessorius	0.3699	0.4637	0.4632	0.4653
Regio retrobulbaris	0.1727	0.1954	0.1955	0.2180
Regio praepiriformis	1.3894	1.4515	1.3437	1.3405
Regio periamygdalaris + Nucleus amygdalae	0.8666	0.9641	0.9643	0.9984
Tuberculum olfactorium	0.1988	0.2511	0.2094	0.2008
Basal nuclei + Commissura anterior	0.9989	0.9151	0.9002	1.0107
Limbic structures	5.1298	5.5362	5.5630	4.9521
Septum	0.6052	0.6313	0.5240	0.5657
Septum nuclei	0.4404	0.4649	0.3827	0.4270
Diagonal band	0.1648	0.1676	0.1405	0.1388
Hippocampus	3.6248	3.7723	4.0410	3.4426
Schizocortex	0.8999	1.1326	0.9988	0.9437

The hanging ears observed in many domestic animals are correlated closely to changes of the auditory canal and of the tympanic cavity (Darwin, 1868; Vau, 1936; Herre, 1953; Fleischer, 1970). Herre (1953) investigated the bullae tympanicae of mufflons and domestic sheep. We may conclude from his findings that changes in the shape and internal structure may be linked with functional reductions. On the whole the 21.5% reduction of the midbrain shows that this brain subdivision in domestic sheep amounts to only about 4/5 of the volume found in wild sheep. The reduction rate, however, does not provide any information on the reduction rate of optic, acoustic and neencephalic centres in the midbrain which have certainly undergone changes during domestication, too.

In domestic sheep the volume reduction of the whole diencephalon equals 21.2%. This value is probably, like that of the mesencephalon, correlated closely with reductions of optic and acoustic centres (Kruska, 1972).

1. Telencephalon

According to cytoarchitectonic characteristics the telencephalon of sheep could first be subdivided into neocortex, corpus striatum and allocortex (Stephan, 1951,

Table 6. Relative percentages (RV) of measured brain structures of the domestic sheep brains. Pure brain volume = 100%

	Sk1	Sk2	Sk3	H1	H2	H3
Medulla oblongata	5.1683	6.1486	5.0369	5.8979	5.8666	5.7394
Cerebellum	11.9634	12.4275	12.3610	12.4654	10.8482	12.5498
Mesencephalon	4.5070	4.8856	4.5570	5.7694	5.1068	4.8994
Diencephalon	6.7259	6.6094	7.3173	7.8875	8.1736	7.3015
Telencephalon	71.6344	69.9289	70.7278	67.9809	70.0058	69.5110
Neocortex	59.4617	58.3410	58.2872	54.2909	56.5973	58.2739
Corpus striatum	3.4202	3.1963	3.5512	4.0755	4.4110	3.2619
Allocortex	8.7534	8.3906	8.8894	9.6145	8.9965	7.9764
Olfactory Allocortex	4.3829	4.0724	4.1933	4.5941	4.2987	3.9937
Bulbus olfactorius + B. olf. accessorius	0.5594	0.4954	0.5538	0.5528	0.5548	0.4638
Regio retrobulbaris	0.1516	0.1366	0.1622	0.1671	0.2079	0.1477
Regio praepiriformis	1.5620	1.3723	1.5001	1.4249	1.2516	1.4089
Regio periamygdalaris + Nucleus amygdalae	1.0204	0.9689	0.9483	0.9225	0.8515	0.8598
Tuberculum olfactorium	0.2562	0.3050	0.2755	0.2743	0.2434	0.2789
Basal nuclei + Commissura anterior	0.8333	0.7932	0.7544	1.2524	1.1895	0.8357
Limbic structures	4.3705	4.3192	4.6962	5.0215	4.6988	3.9816
Septum	0.6339	0.5491	0.5797	0.7221	0.6613	0.6508
Septum nuclei	0.4796	0.3998	0.4261	0.5410	0.4651	0.4474
Diagonal band	0.1543	0.1493	0.1536	0.1811	0.1961	0.2046
Hippocampus	2.8058	2.7520	2.9072	3.3373	3.1072	2.4918
Schizocortex	0.9308	1.0172	1.2093	0.9621	0.9303	0.8390

1954a, b, 1964; Kruska, 1970). For domestic sheep the calculations of volumes (Table 6) resulted in the following reductions of the three large telencephalic areas (Table 7 and Fig. 4):

Neocortex 26.2%, corpus striatum 20.8%, allocortex 29.3%.

The reduction of neocortex and allocortex is clearly more pronounced than in the total brain where 23.9% is observed. Furthermore it is quite obvious that in these two components the neocortex as the highest centre of integration in mammals is reduced less than the allocortex. We find here a difference to other results so far reported for other domestic animals.

a) Neocortex. The subdivisions of the neocortex in wild and domestic sheep will be examined in detail in a later study. In this paper reductions in the total neocortex of domestic animals so far reported are compared. The following data (collected from the same references as in Table 9) are available:

Albino rat 10.9%, domestic sheep 26.2 %, poodle 35.1%, domestic pig 37.3%, ferret 39.1%.

In all cases the reduction found in the neocortex surpassed that of the telencephalon (see Table 9).

Table 7. Geometric mean values (GM), geometric variability coefficients (GVC) and relative values (RV) of body and brain weights (g), brain and brain subdivision volumes (mm^3) of wild and domestic sheep

	Wild sheep ($n=4$)			Domestic sheep ($n=6$)		
	GM	GVC	RV	GM	GVC	RV
Body weight	28605			40036		
Brain weight	132.02	1.0		111.43	9.2	
Total brain volume	128399	1.0		107558	9.2	
Pure brain volume	123180	1.7		101867	8.8	
Medulla oblongata	6094	9.6	4.9472	5734	10.0	5.6289
Cerebellum	13456	3.9	10.9239	12312	9.6	12.0864
Mesencephalon	5899	6.4	4.7889	5030	7.6	4.9378
Diencephalon	8702	4.3	7.0645	7451	7.3	7.3144
Telencephalon	88998	1.5	72.2504	71263	10.4	69.9569
Neocortex	73079	2.0	59.3270	58592	11.4	57.5181
Corpus striatum	4294	6.3	3.4860	3695	14.2	3.6273
Allocortex	11613	2.6	9.4277	8919	11.0	8.7555
Olfactory Allocortex	5094	2.2	4.1354	4331	10.3	4.2516
Bulbus olfactorius + B. olf. accessorius	540	10.2	0.4384	539	14.4	0.5291
Regio retrobulbaris	240	8.7	0.1948	164	16.3	0.1610
Regio praepiriformis	1701	4.4	1.3809	1443	13.8	1.4166
Regio periamygdalaris + Nucleus amygdalae	1167	4.8	0.9474	944	15.6	0.9267
Tuberculum olfactorium	264	10.8	0.2143	277	12.1	0.2719
Basal nuclei + Commissura anterior	1176	7.3	0.9547	941	21.6	0.9238
Limbic structures	6515	5.3	5.2890	4586	12.2	4.5020
Septum	715	9.4	0.5805	642	8.0	0.6302
Septum nuclei	527	9.3	0.4278	466	10.7	0.4575
Diagonal band	188	11.7	0.1526	175	8.7	0.1718
Hippocampus	4575	5.8	3.7141	2931	13.3	2.8773
Schizocortex	1220	9.4	0.9904	993	19.1	0.9748

The percentage of the neocortex within the telencephalon found in some species which have been domesticated is given below:

Wild rats 55.0%, polecat (mustelis) 70.3%, wild boar 77.9%, mufflon 82.1%, wolf 89.3%.

The relatively small percentage of the neocortex within the telencephalon of rats may be regarded as an expression of a low evolutionary level of the rodents if compared with ungulates and carnivores. This difference may be illustrated by determining the participation of the neocortex in the reduction of the total brain which is independent from the body size:

Albino rat 42.1%, ferret 59.2%, domestic pig 63.3%, domestic sheep 65.1%, poodle 74.8%.

Table 8. Percentage reductions of brain weights, brain volumes and volumes of brain subdivisions from wild to domestic sheep. Calculated for the allometric exponents: 0.20, 0.25 and 0.30

	$a = 0.20$	$a = 0.25$	$a = 0.30$
Brain weight	22.48	23.88	25.24
Total brain volume	22.48	23.88	25.24
Pure brain volume	23.47	24.85	26.20
Medulla oblongata	11.80	13.39	14.93
Cerebellum	14.23	15.78	17.28
Mesencephalon	20.07	21.51	22.92
Diencephalon	19.74	21.18	22.60
Telencephalon	24.94	26.30	27.61
Neocortex	24.84	26.20	27.52
Corpus striatum	19.34	20.79	22.21
Allocortex	28.01	29.30	30.57
Olfactory Allocortex	20.30	21.74	23.14
Bulbus olfactorius + B. olf. accessorius	6.44	8.13	9.77
Regio retrobulbaris	35.93	37.09	38.21
Regio praepiriformis	20.48	21.91	23.31
Regio periamygdalaris + Nucleus amygdalae	24.17	25.54	26.87
Tuberculum olfactorium	1.64	3.42	5.15
Basal nuclei + Commissura anterior	24.99	26.34	27.66
Limbic structures	34.01	35.21	36.36
Septum	15.84	17.36	18.84
Septum nuclei	17.10	18.60	20.05
Diagonal band	12.73	14.30	15.83
Hippocampus	39.95	41.01	42.08
Schizocortex	23.70	25.08	26.42

Table 9. Reduction (in percent) of brain volumes in domestic animals

	Total brain	Telencephalon	Diencephalon	Mesencephalon	Cerebellum	Medulla obl.
Albino rat (Graubohm, 1971; acc. to Herre, Röhrs, 1973)	8.7	10.1	6.3	3.8	7.5	3.6
Domestic sheep	23.9	26.3	21.2	21.5	15.8	13.4
Domestic pig (Kruska, 1972)	32.3	36.2	32.8	28.3	25.9	26.6
Ferret (acc. to Schumacher, 1963)	32.7	36.8	32.7	17.4	23.9	16.0
Poodle (acc. to Schleifenbaum, 1973)	33.2	34.4	28.7	16.4	30.2	22.4

Table 10. Participation (in percent) of the reduced volumes of brain subdivisions in the total brain reduction. Total reduction = 100%

	Telencephalon	Diencephalon	Mesencephalon	Cerebellum	Medulla obl.
Albino rat	70.9	7.3	3.1	13.0	5.6
Domestic sheep	79.6	6.3	4.3	7.2	2.8
Domestic pig	78.9	6.8	3.4	9.1	4.6
Ferret	79.2	6.7	1.8	10.3	2.9
Poodle	81.9	3.9	1.1	9.5	2.2

If the changes of total brain weight in domestication are compared one can observe a dependence of the neocortical development. That is the total reduction rate is low in more primitive domestic mammals and more pronounced in species with a more highly developed neocortex.

"Obviously the conditions of domestication do not influence the brain of primitive mammals as much as the brain of highly developed species. The fact that it is just the phylogenetically youngest and most developed part of the mammal brain which may change so much under domestication, is a plead in favour of the theory that there exist important relations between the variability of the neocortex and its phylogenetic development" (Herre and Röhrs, 1973, p. 156).

b) Corpus Striatum. The parts of the nucleus caudatus and the putamen form the corpus striatum. It includes the major part of the basal ganglia. The nucleus accumbens forms the basal portion of the caudatum and is continued medially with the septum (Kuhlenbeck, 1927; Szteyn, 1965). This is why the main part of the nucleus accumbens was measured together with the striatum, while part of it—a rest which cannot be determined exactly—is measured together with the septum. Because of its position the globus pallidus is often described in connexion with the basal ganglia. It is, however, of thalamic origin and is therefore part of the diencephalon (Spatz, 1949). The question if the claustrum originates from the cortex or the subcortex has so far not been solved definitely (De Vries, 1910; Filimonoff, 1966). In accordance with Kruska (1970) the claustrum has been associated with the neocortex. The amygdaloid complex shows fibre connexions with the cortex centres of the telencephalon and with the thalamic areas (Lammers, 1972). For technical reasons this nucleus complex has been measured together with the structures of the allocortex.

Comparative investigations of volume reductions of the corpus striatum of wild and domestic animals have already been carried out on rats (Graubohm, 1972), pigs (Kruska, 1970) and poodles (Schleifenbaum, 1973). The results obtained from rats (10.2%), pigs (26.4%), and poodles (26.1%) show, just as those from sheep (20.8%), in each case considerable reductions of this brain subdivision.

Particularly instructive is the comparison of the percentage striatum reduction in the reduction of the total brains of the following animals:

Albino rat 5.5%, domestic sheep 3.0%, domestic pig pig 2.9%, poodle 2.4%.

These results could indicate a considerable reduction of the extrapyramidal functions of albino rats. One reason for the large decrease might possibly be the keeping in cages of the albino rats which among other factors has contributed to a considerable change of body exercise compared to the conditions of wild life. Hence it might be conceivable that the conditions of captivity of domestic animals may influence the development of certain brain subdivisions. Sheep and pigs are subject to similar conditions of captivity. Similar reduction values of the striatum equalling about 3% could prove this hypothesis as to motor activity.

c) Allocortex. The classification of allocortical areas as suggested by Stephan (1964)—in accordance with Filimonoff (1947), Ariens Kappers (1909), Vogt (1927) and other authors is given below:

	A. Allocortex primitivus	I. Palaeocortex
		II. Archicortex = Hippocampus
Allocortex		
	B. Periallocortex	III. Peripalaeocortex
		IV. Periarchicortex

The allocortex has a simpler structure than the isocortex. The cell layers are less distinct and the typical 6-layer pattern of the isocortex can only be recognized partly. The allocortex can be divided into the following regions (Stephan, 1964):

I. 1. Bulbus olfactorius (Bol) 2. Bulbus accessorius (Bac) 2. Regio retrobulbaris (Rb) 4. Regio praepiriformis (Prpi) 5. Regio periamygdalaris (Pam) 6. Tuberculum olfactorium (Tol) 7. Regio corticalis septi (S) 8. Regio diagonalis = Diagonal Band of Broca (DB).

II. 9. Subiculum 10. Cornu ammonis 11. Fascia dentata 12. Taenia tecta, Rose.

III. 13. Formatio mesocorticalis insularis.

IV. 14. Regio entorhinalis 15. Regio praesubicularis 16. Regio retrosplenialis 17. Regio infraradia ventralis 18. Regio subgenualis posterior.

Stephan (1960b, 1961, 1964) and Hassler (1964) distinguish in the allocortex between two functional systems which partly correspond to the cytoarchitectonical subdivision. The first is an olfactory system, largely depending on sense organs, the second a higher control centre, independent from sensory input (Stephan, 1961), the so-called "limbic system". The olfactory system is connected with centres of the palaeocortex. Stephan (1960b) distinguishes between primary olfactory (1–2) and secondary olfactory centres (3–6). The limbic structures, on the contrary, are mainly archicortical and periarchicortical; to these belong the areas no. 9–12 of the archicortex, 14–18 of the periarchicortex and probably even the septal structures of the palaeocortex (Stephan, 1964, Stephan and Andy, 1962, 1970). The areas no. 14–15 were combined by Stephan (1960b) according to Rose (1926) and called Schizocortex.

Olfactory primary centres (Bol+Bac), the hippocampus (Hi) and the schizocortex (Sch) have not been subdivided further in this study. In all cases the strongly stained mitral cell layer (IV) was regarded as the outer limit of the bulbus, as it was not always possible to prepare the bulbus without damage. The volume of the secondary olfactory centres contains the amygdaloid complex, too. By the cortex structures of the amygdala which are called "Regio pariamygdalaris"

(Rose, 1927) this nucleus region is interconnected with the olfactory system. We should, however, point out again, that impulses from other sensory systems reach the amygdala, too (Lammers, 1972). The structures of the overlapping regions between is palaeo-, and archicortex (proisocortex, peripalaeocortex, periarchicortex) cannot always be defined clearly according to cytoarchitectonical and functional considerations (Stephan, 1964). This is why the areas 13, 16, 17 and 18 have not been subdivided but measured together with the isocortex. Altogether 10 allocortical structures could be circumscribed, and their volume changes in domestic sheep compared to wild sheep could be determined. The changes are given in Table 7. Fig. 4 is intended to illustrate the degree of the changes, and Fig. 5 is intended to show the dimensions of superficial allocortex structures of a wild sheep in medial and ventral view.

All measured allocortex segments put together decrease in sheep under domestication by 29.3% in volume. This value surpasses the reduction value of the neocortex by 3.1%.

The investigations of Kruska (1970), and Kruska and Stephan (1973) concerning pigs' brains, of Schleifenbaum (1973) concerning poodles' brains and of Graubohm (1971) concerning rats' brains, provide different results, in all cases there have been calculated larger reduction values of the neocortex compared to those of the allocortex. In pigs the allocortex has been reduced by 35.1%, in poodles by 29% and in rats by 8.9%.

The olfactory allocortex decreased in domestic sheep—compared to mufflons—by 21.7%. Compared to the reduction of the neocortex this value is even smaller. For the olfactory primary centres, i.e. Bol and Bac, the reduction amounts to 8.13%. Next to the tuberculum olfactorium with a reduction of 3.42% the bulbus shows the smallest reduction value in the total brain of sheep. Much higher, on the contrary, are the reductions of other areas of the olfactory allocortex. In domestic sheep the regio retrobulbaris has a volume which is by 37.1%, the basal nuclei by 26.3%, the regio periamygdalaris with the nucleus amygdalae by 25.5% and the regio praepiriformis has a volume which is by 21.9% smaller than in wild sheep. The secondary olfactory centres are therefore subject to larger reductions under domestication than the primary centres—except for the tuberculum olfactorium. In these volume calculations for secondary centres there have been measured relatively large structures which are not exclusively used for olfactory functions: the amygdaloid complex and the basal nuclei which have been circumscribed together with parts of the commissura anterior. The larger reductions may therefore be attributed to reductions of other functional systems, too.

Comparable values on reductions of the "rhinencephalon" in pigs (Kruska, 1970; Kruska and Stephan, 1973) and dogs (Schleifenbaum, 1973) have been reported so far. These amount to 21.7% for poodles and to 30.9% for domestic pigs. In a similar way as for sheeps the bulbus olfactorius of domestic pigs shows a smaller reduction—compared with the secondary olfactory centres—of 15.1%. It may generally be assumed for the olfactory allocortex of sheep, pig and dog that this brain subdivision has not reduced so much under domestication as the neocortex.

The structures of the limbic system, however, in domestic sheep show a reduction value (35.2%) which surpasses considerably the reduction of the total brain and the reduction of the neocortex. When the non-olfactory allocortex is subdivided into septum, hippocampus and schizocortex, these areas show the following reductions (in domestic sheep):

Septum 17.36%, schizocortex 25.1%, hippocampus 41%.

The total reduction of the limbic structures of the allocortex is therefore composed of differently dimensioned reductions of sub-areas, just like that of the olfactory allocortex. The reductions of the septum are the smallest, whereas those of the hippocampus are the greatest. In addition to this, the hippocampus has decreased most of all brain subdivisions during domestication. The septum of the sheep was subdivided into septal nuclei and diagonal band, and measured separately. The septal nuclei have reduced by 18.6%, the diagonal band, on the contrary, only by 14.3%.

For the reduction values of the limbic structures in domestic sheep there may be used comparative values from domestic pigs and poodles, as well. The overall decrease of the non-olfactory allocortex which for pigs amounts to 39.5% and for poodles equals (36.2%, shows similar values in domestic sheep. In accordance with the values found in domestic sheep the hippocampus of pigs (42.4%) and poodles (38.4%) undergoes the greatest reduction in the brain of these animals. The schizocortex, however, has decreased much more in domestic pigs (38.4%) than in poodles (25.8%) and domestic sheep (25.1%). The septum of the poodles (29.2%) was reduced more than that of the pigs (25.8%) and sheep (17.4%) and surpasses thereby the reduction of the schizocortex. The topographic thereby the reduction of the schizocortex. The topographic association of the septum with the palaeocortex and its small reductions—comparable to those of the olfactory centres—could be an argument in favour of the hypothesis that this brain part receives more olfactory than non-olfactory impulses. It is a fact that the position of the septum within the non-olfactory allocortex is partly controversial. Stephan (1964), Hassler (1964) and Stephan and Andy (1962, 1970) associate the septum with the limbic system. Phylogenetic (Johnston, 1913; Stephan, 1961, 1964; Stephan and Andy, 1962, 1970) and functional observations (Hassler, 1964; Covian, 1967) are in favour of this view.

III. Volumetric Comparison of the Brains of North German Moorland Sheep and Blackhead Sheep

North German Moorland Sheep are an old and primitive country sheep breed. The animals are resistant and do not require particular care. Their habitus is similar to that of wild sheep. North German Moorland Sheep are used in many ways for their wool, fur, and meat. Blackhead sheep, on the other hand, differ greatly from the phenotype of the wild species, this may be attributed to the fact that they have been bred exclusively for meat-production and for a high rate of reproduction. They are considered as more exigent and more domesticated than the North German Moorland Sheep. As mentioned above there exist difference in the size of the telencephalon as to shape and fissuration between North German Moorland Sheep and Blackhead Sheep. This is why it is quite interesting to find out if the differences of brain size between the two breeds apply as well to a comparison of volumes independently from body weight.

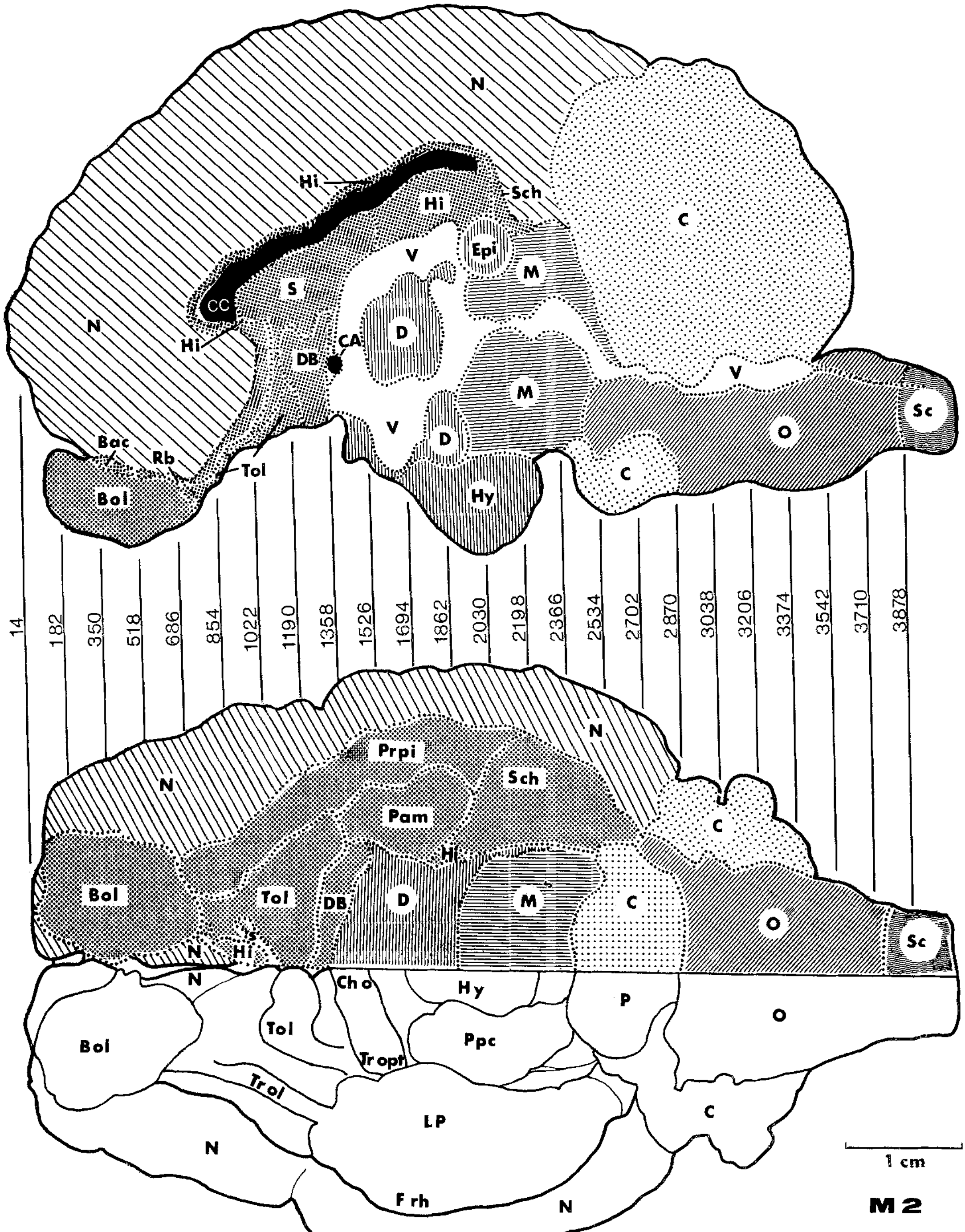

Fig. 5. Reconstructions of the median and ventral view of the brain of a wild sheep by means of photographed serial sections. For abbreviations see p. 273

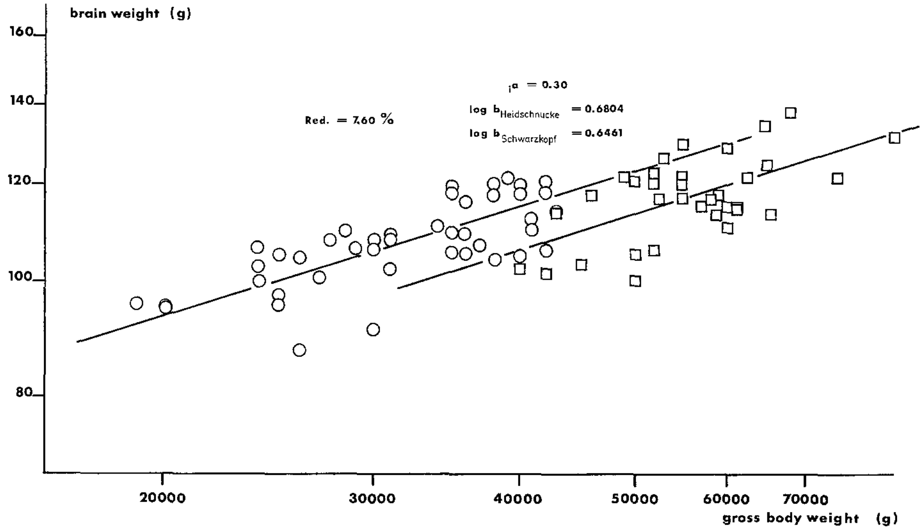

Fig. 6. Relations between brain weight and gross body weight of North German Moorland Sheep (Heidschnucke) and Blackhead Sheep (Schwarzkopf) in a double logarithmic scale. ○ North German Moorland Sheep; □ Blackhead Sheep

For the relation brain to body weight the angles of inclination of the determined allometric line for North German Moorland Sheep and Blackhead Sheep are shown in Table 2. Fig. 6 illustrates the grouping of the pairs of values in a double logarithmic scale. The inclination angles of both lines do not differ significantly from each other so that we may determine a common allometric exponent for both. It equals $a = 0.2955$ (Table 2). The main axes of the ellipses have been proved to differ in their position from each other ($P = 0.01$). Hence the following allometric equations are valid:

$$2.0294 = 0.6804 - 0.2955 \cdot 4.4966 \quad \text{(North German Moorland Sheep)}$$
$$2.0697 = 0.6461 - 0.2955 \cdot 4.7453 \quad \text{(Blackhead Sheep)}$$

Animals of both breeds with the same body weight differ in their brain weight; the brain weight (and volume) of the Blackhead Sheep is by 7.6% smaller than that of the North German Moorland Sheep.

A volumetric comparison shows that all brain parts listed in Table 11 have lower values in Blackhead Sheep than in North German Moorland Sheep. The volume difference are not equally dimensioned. They vary between 3.8% in the neocortex and 19.5% in the corpus striatum. The following percentage differences of the five brain subdivisions of North German Moorland Sheep to Blackhead Sheep could be determined:

Telencephalon 5.5%, diencephalon 11.3%, mesencephalon 18.2%, cerebellum 5.1%, medulla oblongata 12.0%.

A significant relation between these data and the reduction value found in the comparison of wild sheep to domestic sheep cannot be recognized. If the large variability coefficients of the absolute volumes of the brain subdivisions in

Table 11. Geometric mean values (GM), geometric variability coefficients (GVC), relative values (RV) and differences in percentage of brain subdivisions comparing North German Moorland Sheep (Heidschnucke) to Blackhead Sheep (Schwarzkopf)

	Heidschnucke ($n = 3$)			Schwarzkopf ($n = 3$)			Reductions ($_i$a = 0.30)
	GM	GVC	RV	GM	GVC	RV	
Body weight	29920	22.8		53572	23.1		
Brain weight	103.8	4.9		119.6	4.8		7.60
Total brain volume	100253	4.9		115394	4.8		7.60
Pure brain volume	95310	5.7		108875	4.1		8.30
Medulla oblongata	5561	6.6	5.8346	6049	13.5	5.5559	12.01
Cerebellum	11365	2.9	11.9242	13337	3.1	12.2498	5.08
Mesencephalon	5000	10.2	5.2460	5059	6.3	4.6466	18.16
Diencephalon	7414	11.5	7.7788	7488	2.8	6.8776	18.30
Telencephalon	65918	6.7	69.1617	77041	4.7	70.7610	5.46
Neocortex	53721	6.5	56.3645	63904	5.1	58.6948	3.78
Corpus striatum	3703	22.7	3.8852	3687	4.9	3.3865	19.46
Allocortex	8422	13.3	8.8364	9445	3.9	8.6751	9.29
Olfactory Allocortex	4088	10.3	4.2892	4588	6.8	4.2140	9.22
Limbic Structures	4333	16.3	4.5462	4855	2.1	4.4592	9.37

domestic sheep in general (Table 7) and those of the North German Moorland Sheep in particular (Table 11) are taken into account, the size differences of the brain subdivisions of Blackhead Sheep compared to North German Moorland Sheep may possibly be attributed to the large dispersion rate of the volumes. A particularly high variability of organ sizes in domestic animals has been observed by several authors (see Herre and Röhrs, 1971, 1973, and other authors). This is why it seems adequate to approve of the calculated differences of brain volume and of the volume of brain subdivisions—independent from body weight—between North German Moorland Sheep and Blackhead Sheep only with a reservation to the variability. On the whole this result may only provide an indication to possible differences of brain size in different domestic breeds of the same species. Up to now there is no clear evidence of diverging brain reduction due to different keeping conditions of domestic animals (cf. Gaudlitz, 1971). In spite of this in future brain investigations on wild and domestic animals the breed of the domestic animal should be stated in order to get informations on possible characteristics of brain structure found in a particular breed (cf. Schleifenbaum, 1973).

D. Discussion

The process of domestication is of high interest to the Zoologist, as it has to be considered as the greatest experiment which man carried out with animals (Herre and Röhrs, 1971, 1973). For many generations animals of various wild species have been submitted to conditions which changed their natural way of life to a great extent. By isolating small groups of wild populations—sexually separated from the wild species—, animals which were considered suitable for certain purposes were bred to form large herds. During this time the social structure, the natural conditions of reproduction, the natural selection pressure, alimentation and environment of the animals were changed considerably. Under these special conditions of domestication there developed great changes in appearance, organ structure and function, as well as in behaviour of the domestic animals as compared to those of the wild species.

One object of the domestic animals biology is the quantitative determination of such changes which are caused by domestication. Particularly interesting are the changes in the brain. By means of allometric analyses biologists have been able to prove that domestic animals show—in some cases considerable—reductions of the brain compared to that of their wild species (Herre and Röhrs, 1971, 1973). According to present opinions the extent of the brain reductions is determined by several factors.

1. Evolution, and closely connected with it the cerebralization of the animals under domestication possibly influence the "species-typical" reaction to the changed conditions of life. Thus in the list on page it is obvious that the rodents and lagomorphs whose brains are on a low evolutionary level within the mammals, do not show any or only low brain reductions during domestication. In contrast to this the more highly evolved brains of the ungulates and carnivores are much moe reduced in the domestic state. Within these groups, however, there exist considerable differences in artiodactyla (llama, alpaca), perissodactyla (horse, ass), and carnivores (cat) compared to the "normally high" reduction values expected. In a comparison of the total brain reductions determined so far for artiodactyla, domestic sheep with a brain reduction of 24% hold a middle position between llama/alpha (19%) and domestic pigs (34%). It is difficult to interpret these differences. Further it is not very easy to find out to what extent the reductions are influenced by the factors discussed here (cf. Kruska, 1973).

2. Specializations of the brain mainly occur in those brain centres which are correlated to the sensory organs nose, ear and eye. Under the special conditions of domestication the functional strain of these organs and therefore that of the correlated brain centres, too, is reduced to some extent. The hight of the developmental level of the brain centres, i.e. the extent of specialization, might be responsible for a corresponding large reduction in the relatively monotonous environment of the domestic animals. This hypothesis cannot be proved by cytoarchitectonic investigations. It is a fact, however, that the centres for olfactory, acoustic and optic functions in the brain of domestic animals have undergone changes of size under domestication, if compared with their wild species (Herre and Röhrs, 1971, 1973, and other authors). Kruska (1972) and Kruska and Stephan (1973) were able to demonstrate considerable reductions of optic, olfactory and libic centres in the brains of domestic pigs. The comparison between the brains of wild and domestic sheep, too, shows reductions of the "olfactory brain" and the limbic system of domestic sheep. The differently dimensioned reduction values within the olfactory and non-olfactory allocortex indicate that the overall functions of these regions result from several components linked to morphologically defined structures. Direct information on the functions in the different structural units may only be obtained by means of electro-physiological and experimental-anatomical experiments, the results of which, however, are still not very conclusive and partly contradictory. Comparative neuro-anatomic studies

may provide indications on the possible functions of certain areas. The publications of Stephan (1960b, 1961, 1966, 1967) and Stephan and Andy (1970) show, for example, that in the phylogenetic development of basal insectivores to higher primates the olfactory centres are reduced to a great extent whereas the limbic parts of the telencephalon do not decrease but increase. Hence the limbic structures do not participate substantially in the olfactory function of the brain of these animals. Another example taken from the comparative neuro-anatomy is intended as well to illustrate the relation between size and function of a structure. In aquatic animals the olfactory centres are, as a rule, reduced; the limbic areas, on the contrary, are well developed (Filimonoff, 1965). According to these findings a close functional relation between the olfactory and limbic system has to be denied.

The reductions of the allocortex therefore indicate considerable reductions of some activities controlled by the central nervous system in domestic sheep compared to wild sheep. In addition to this the different rates of reduction in different areas show that the functional changes have not influenced the brain structures in a uniform way. The quantitative data in olfactory allocortex seem to indicate a reduction of the sense of smell. For the wild sheep the olfacotry sense is—next to the optic system—of considerable importance for the general orientation, recognition of the enemy, food search and selection. In addition this sense plays an important role in social life (recognition and sexual behaviour) (Türcke and Schmincke, 1965; Altmann, 1970; Geist, 1971).
— For domestic sheep the environmental conditions and the social conditions have changed considerably compared to the natural life conditions of the wild species. It might be conceivable that the sense of smell in domestic sheep is less used for recognition of the enemy, general orientation and food search than in wild sheep, as man has taken over most of these cares. Social relations of domestic sheep have changed in so far as the small groups of the wild state have been replaced by large herds in the domestic state. Olfactory activities which contribute to individual recognition and to reproductive behaviour probably keep their functional importance even in larger herds. In contrast with the olfactory system the limbic system represents a superior control centre which is influenced by all sorts of sensory input (Hassler, 1964). It contributes to the control of a number of vegetative functions, sexual behaviour, food uptake, memory and extrapyramidal-motor functions. Its manifold functions are localized in complex structures. The hippocampus has a central position in this system (Stephan, 1961; Hassler, 1964). "Some of the supposed functions or subfunctions are: general cerebral activation (Herrick, 1933), control of emotions, of affective behaviour (Papez, 1937), control of the normal function of the neocortex (Grünthal, 1947), self-preservation and preservation of species (Maclean, 1958) and the registration of time (Hassler, 1959)" (Stephan, 1961, p. 46).

It may well be assumed that the high reduction rate of the hippocampus in domestic sheep which surpasses by far all areas investigated here, should be brought into relation with some of the above-mentioned functions which were probably reduced to a great extent during domestication. Suppression of flight behaviour and agression instinct must be regarded as essential conditions for a successful domestication.

The general alertness characteristic for wild animals has decreased considerably in domestic animals. The struggle for existence is less hard in the domestic state. Domestic animals live in a "relaxed sphere" (Lorenz, 1959). According to Lorenz the endogenous inproduction of stimulation of some instinct movements is subject to quantitative changes which are due to domestication. Reductions of warning behaviour and flight and defense acts have also to be regarded within this context. The "warning whistle" characteristic for mufflons has been lost in domestic sheep.

There exist, however, hypertrophic behaviour patterns in domestic animals, compared to that of their wild species. In this respect one should mention particularly some aspect of reproductive behaviour. Sexual activity in domestic animals is often increased. The animals are earlier sexually mature and the sequence of estrus cycles is increased, so that the reproduction rate goes up (Herre and Röhrs, 1971, 1973). If within the limbic system there are structures which might be correlated with sexual behaviour these differ possibly from the large reduction of the total system. In this respect the low reduction of the septum are of interest. In fact, some authors have discussed—among other relations—close relations of the septum to the reproductive system (Harterius after Andy and Stephan, 1964). Kruska and Stephan (1973) determined for the septum of pigs the smallest values, too, within the non-olfactory allocortex.

3. It is conceivable that diverging methods of captivity and breeding aims (e.g. animals kept for meat or wool production, pets, working animals and those kept for other reasons) influence the central nervous system in a highly differentiated way. Taking two breeds of domestic sheep as an example there could be shown a small but significant difference in the size of the total brain. The investigations of the brain subdivisions taken separately resulted in small differences of the brain subdivision volumes in Blackhead Sheep compared to that of German Moorland Sheep, too. Because of the considerable dispersion rate of the volume values, however, this result may not be considered as representative.

4. Changes in the domestic state may be modificatory or genetic. Many domestication phenomena are interpreted as modifications (Herre and Röhrs, 1971, 1973). The brain, too, adapts himself to the more or less changed environmental conditions. We may, however, presume that at a given moment which cannot be determined exactly the size of the brain has been fixed genetically according to the characteristic conditions of domestication. Crossbreeding of wolf and poodle (Herre, 1966; Weidemann, 1970) show that changes of brain weight during domestication may not be interpreted simply as modifications. On the other hand a high modificability can be observed in captivity, too (Klatt, 1932; Stephan, 1954b; Stockhaus, 1965; Herre and Röhrs, 1971, 1973). Probably the modificatory brain changes of the first generation in captivity have a favourable effect on the life in the domestic state, comparable to the first generations of domestic animals. It is conceivable that in the course of many generations the altered conditions of environment, selection, reproduction and nutrition had not only modificatory influences but also changed the genetic structure of domestic animals (Herre and Röhrs, 1971, 1973).

On the whole the changes of brain size in domestic sheep have shown that the divergent reductions of certain brain subdivisions may be functionally correlated to changes of the environmental conditions.

Comparative-cytoarchitectonic investigations, however, may only provide indications on the function of certain structures. More detailed electro-physiological or experimental-anatomical investigations in wild and domestic animals are desirable and could help to answer many open questions.

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